

MOBILE & WEB APP — SOLUTION PROPOSAL

A Complete Learning Platform for Al Qalam.

A purpose-built mobile and web platform for Al Qalam Learning — bringing students, teachers, and administrators onto one unified system. Built for speed of delivery, low operating cost, and the dignity an Islamic learning institute deserves.



Built for the mission of "Nurturing Upon Salafi Tarbiyah" — founded by dedicated Salafi sisters. This proposal carries Al Qalam's four programmes — *Academic with Islamic Course, Madrasa with Hifdh, Hifdh Only, and Individual Tuition* — into one digital platform: a Mobile App (iOS + Android), an Admin & Teacher Web Portal, and Desktop access for students. The goal is the same as yours — to prepare students for both this world and the Hereafter, through technology that respects the depth of what is being taught.

Client	Al Qalam Learning
Prepared by	Shabeeb K · CodeOx · +966 53 571 6437 · shabeeb.k@code-ox.com
Our Portfolio	my-docs-sigma-beige.vercel.app/portfolio
Scope	Mobile App (iOS + Android) · Web Portal (Admin + Student) · Backend
Roles Supported	Student · Teacher · Admin
Languages	English & Malayalam (bilingual UI)
Phase 1 Features	52+ features across 15 modules
Timeline	4 Months · 4 Build Phases · Production-ready delivery
Payment Terms	40% Advance · 40% on UAT Submission · 20% on Go-Live
Investment	₹ 2,50,000 (All-inclusive, Phase 1)
Document Ref	AQL-PROP-2026-01 · Version 1.0

01 Project Summary

Al Qalam Learning becomes a fully digital school — students view their schedule, homework, attendance, and marks from a single mobile app. Teachers mark attendance, upload materials, and communicate directly with students. Administrators run admissions, calendars, reports, and broadcasts from a dedicated web console. All three roles share a single source of truth, accessible on phone, tablet, and desktop.

3

APPS DELIVERED

52+

PHASE 1 FEATURES

15

FUNCTIONAL MODULES

4

MONTHS TO LAUNCH

02 The Three Applications We Will Build



Mobile App — iOS & Android
React Native · single codebase · tablet-ready
Users: Students & Teachers
The daily-use app. Students see homework, attendance, schedule, materials, marks, certificates, and chat with their teacher. Teachers mark attendance, upload work, enter marks, send announcements — all from the phone.



Admin & Teacher Web Portal
Next.js · runs on any browser · responsive
Users: School Admin, Principal, Teachers
The control room. Manage students, teachers, classes, admissions, academic calendar. Enter exam marks, generate report cards, broadcast announcements, review attendance reports, and approve leave requests at scale.



Student Web Portal — Desktop
Same web platform · student-facing routes
Users: Students (browser access)
For students who prefer a computer — same data as the mobile app, optimised for desktop. Notes, homework submission, schedule, certificates, report cards, and chat — accessible from any laptop or PC browser.



Shared Cloud Backend
Node.js · Express · MongoDB · Cloud VPS
Powers: All three apps
One source of truth. Battle-tested Node.js API on a dedicated cloud VPS, with MongoDB powering flexible data storage for students, classes, marks, content and chat history. Real-time delivery via Socket.io.

03 How the Platform Maps to Your 4 Programmes

Academic with Islamic Course
Daily homework & submissions · Attendance & marks · Notes & worksheets · Islamic study resources · Exam timetable & report cards · Bilingual EN/ML UI.

Madrasa with Hifdh Course
Madrasa schedule alongside academic timetable · Notes for both streams · Teacher feedback per subject · Workshop & Musabaqa registrations · Achievement badges & certificates.

Hifdh Only Course
Dedicated Hifdh schedule · Ustadh-student 1:1 chat for revision · Attendance for Hifdh sessions · Event reminders & holiday notices · Certificate downloads.

Individual Tuition
One-on-one tutor-student thread · Personal homework upload & feedback · Custom timetable per student · Private notes & worksheets · Push reminders for sessions.

04 Design Phase — What You Get Before a Line of Code Is Written

DESIGNED IN FIGMA All design assets delivered as a Figma file you own and can edit.

The design phase runs through the first 2–3 weeks of the project. Nothing gets coded until you sign off on the look, feel, and flow. You see the entire app — every screen, every interaction — as a clickable Figma prototype before development begins.



Visual Design System

Brand colors, typography, spacing, iconography & component library — the foundation every screen builds from.



Wireframes

Low-fidelity structure of every key screen — student, teacher, admin. Signed off before pixel-pushing begins.



High-Fidelity Mockups

Pixel-perfect Figma frames for mobile, tablet & desktop. Bilingual (EN + ML) layouts included.



Interactive Prototype

A clickable Figma prototype — you tap through the entire app and feel it before a single line of code.



Component Library & Style Guide

Reusable Figma components & documented style guide — handed off to development with zero ambiguity.



Two Revision Rounds

Feedback cycles built into the design phase — we iterate with you until the look & flow are signed off.

SAMPLE COLOUR PALETTE



Final palette chosen with the school during design week 1.

TYPOGRAPHY

Assalamu Alaikum

Clear, friendly type built to read well in both English and Malayalam screens.

Inter · 400 / 600 / 800

COMPONENTS

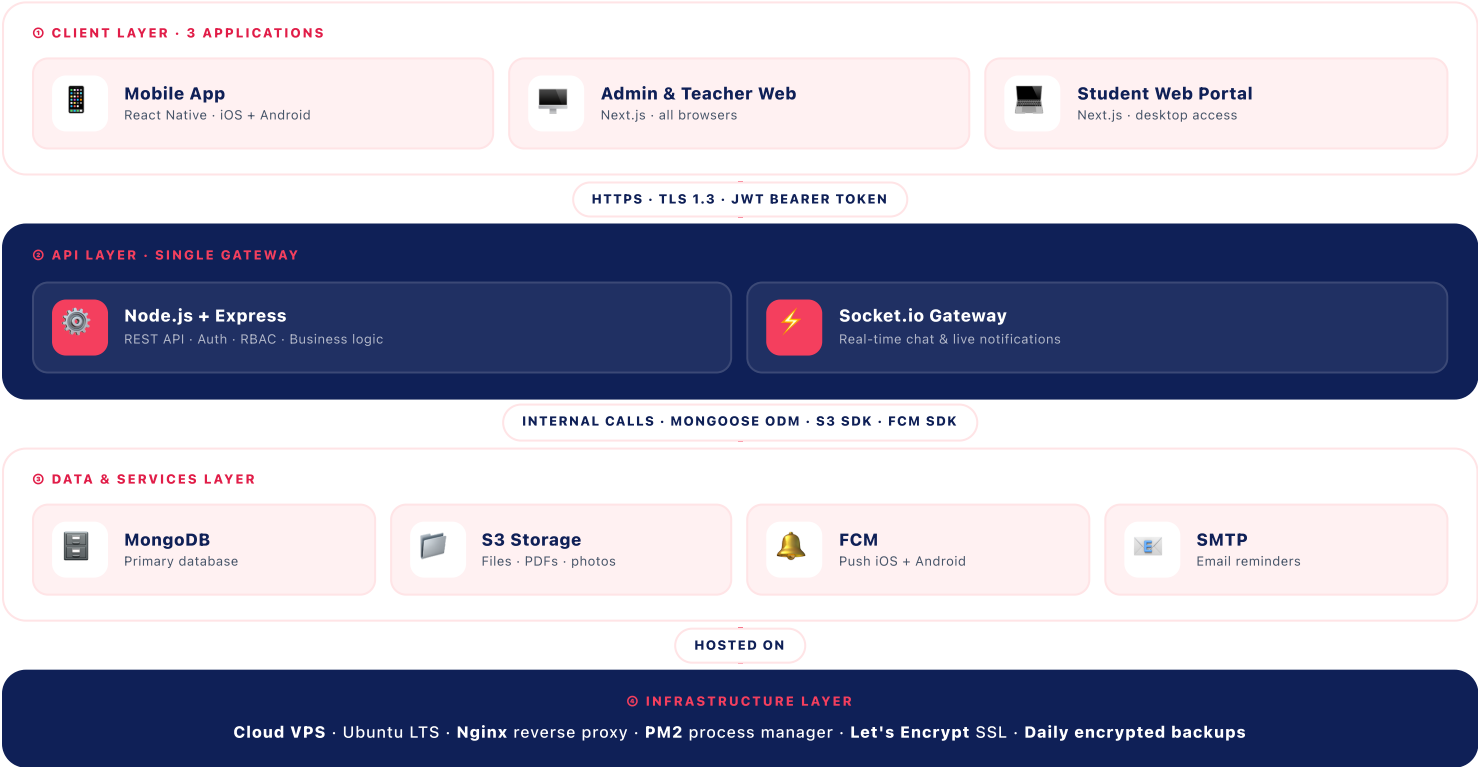


05 Technology Stack

Layer	Technology	Why this choice
Mobile	React Native (Expo)	One codebase for iOS + Android — half the build cost. Over-the-air updates without re-submitting to app stores.
Web Portals	Next.js 16	Industry-leading framework. Server-rendered, fast load times, modern App Router, smooth on any device.
Backend API	Node.js + Express	Mainstream, battle-tested REST API stack. High concurrency, asynchronous, easy to extend.
Database	MongoDB	Flexible document store — perfect fit for school content (notes, materials, mixed data shapes). Scales smoothly from one school to many.
Authentication	JWT + bcrypt	Industry-standard token-based auth. Role-based access for Student / Teacher / Admin built in.
File Storage	VPS + S3-compatible storage	Cloud object storage (DigitalOcean Spaces / Backblaze B2 / Cloudflare R2) — pennies per GB for notes, photos, PDFs, certificates.
Push Notifications	Firebase Cloud Messaging	Free, supports both Android (FCM) and iOS (APNs) from a single API. The standard choice.
Chat (1:1 Student–Teacher)	Socket.io	Real-time bidirectional messaging over the same Node.js backend. Mature, proven library.
Multi-language	react-i18next / next-intl	English & Malayalam from day one. Additional languages can be added later without rebuilding.
Hosting	Dedicated Cloud VPS	A single, high-performance virtual private server with full root access. Predictable monthly cost — no surprise bills, no usage-based pricing traps.

06 System Architecture — Every Layer, End to End

Four clean tiers — clients, API, data & services, and infrastructure. Every request flows through the same authenticated path. No hidden services, no black boxes, no vendor lock-in.



07 Security & Data Protection

Security is built in at every layer — from how passwords are stored to how data is backed up. Six controls, all standard practice for production systems handling student and family data.



JWT Authentication

Industry-standard token-based auth. Short-lived access tokens, secure refresh flow.



Role-Based Access Control

Student / Teacher / Admin see only what their role allows — enforced at the API.



bcrypt Password Hashing

Passwords never stored in plain text. One-way hashed with adaptive cost factor.



HTTPS Everywhere

TLS 1.3 across all clients & APIs. Auto-renewing Let's Encrypt certificates.



Encrypted Backups

Daily MongoDB snapshots, encrypted at rest, 30-day retention with point-in-time recovery.



Audit Logging

All admin actions logged — who changed what, when. Full accountability trail.

08 Reliability Commitments

99.5%

UPTIME TARGET

Daily

ENCRYPTED BACKUPS

30d

BACKUP RETENTION

24/7

ERROR MONITORING

09 What Students & Teachers See Every Day

Student Dashboard

6 features

Everything a student needs to know — at a glance, the moment they open the app.

AL QALAM · HOME

Assalamu Alaikum,

Ali Khan · Grade 7-A

92%

ATTENDANCE

A+

PERFORMANCE

Next: Tafseer · 10:15 AM

> Profile · Class · Student ID

> Live attendance % & performance card

> Today's schedule + upcoming classes

Notifications

6

Push

Exam alerts

Holidays

Events

Live Class Schedule

3

Weekly timetable

Reminders

Attendance Management

5 features

Teachers mark in one tap. Students & admin see live percentages. Monthly reports auto-generated.

Daily mark

Monthly report

Leave application

Approval flow

Homework & Assignments

5 features

Snap a photo, hit submit. Teacher feedback lands in hours, not days.

MATH WORKSHEET 04

Due tomorrow · 11:59 PM

photo.jpg

worksheet.pdf

SUBMIT ASSIGNMENT

Ustadh Rashid: "Great work, Ali — masha Allah!"

Learning Materials

5

Notes

PDFs

Offline view

Islamic library

Exams & Marks

5

Timetable

Marks entry

Report cards

Teacher Portal

6 features

The whole classroom in one screen — attendance, homework, marks, announcements, materials.

Mark attendance

Assign homework

Upload materials

Enter marks

Announcements

A Day in Ali's Life on Al Qalam

Six touchpoints — one app

7:00 AM

1

Opens the app

Sees today's schedule & morning duas

8:30 AM

2

Marked present

Teacher taps attendance in class

11:00 AM

3

Submits homework

Snap photo of worksheet & uploads

2:00 PM

4

Checks marks

Ustadh entered today's quiz score

4:30 PM

5

Asks a doubt

Chats with teacher about tomorrow's test

7:00 PM

6

Reads notes

PDF downloaded for offline reading

10 Communication, Recognition & School Operations



Student ↔ Teacher Chat

4 features

Doubts cleared the same day. Real-time 1:1 messaging, per subject, with the right teacher.

CHAT · USTADH RASHID · MATH

When is the surprise test?

Wednesday 2 PM in sha Allah

JazakAllah khair ustadh 🙏

Revise chapters 4 & 5



Certificates & Awards

4

PDF certificates

Badges

Awards wall



Workshop Registration

3

Event listing

Sign-up form



Academic Calendar

4 features

School calendar, holidays, exam dates, events & workshops — all in one view. Admin updates push to every student instantly.

Holidays

Exam dates

Events

Auto-sync



Fees — Reminders

2

Push reminders only in Phase 1. *Online payment in Phase 2.*



Support & Help Desk

3

Admin contact

WhatsApp btn



Platform & Extras

5 features

Bilingual English & Malayalam, runs on mobile, tablet and desktop, with secure role-based login for every user type.

EN + ML

Mobile

Tablet

Desktop

Role-based access

11 Phase 2 — Future Upgrades (Quoted Separately)

When you are ready to expand

Phase 2 is a separate engagement, scoped after Phase 1 goes live. It adds the engagement layer and the Islamic study module — keeping your investment phased, predictable, and tied to working features.

MODULE	PHASE 2 FEATURES
Live Classes	Google Meet / Zoom deep-link integration · Join Class button · Recorded class library
Online Exams	Timed online exams · MCQ & short-answer quizzes · Auto-grading
Fees	Fee details · Payment history · Online payment gateway · Downloadable receipts
Parent Portal	Dedicated parent login · Child progress dashboard · Teacher remarks · Multi-child switch
Quran Section	Hifdh tracker · Surah memorization log · Tilawah uploads · Revision schedule — built for your Madrasa with Hifdh & Hifdh Only courses
Islamic Studies	Hadith of the week · Daily Adhkar · Dua collection · Seerah lessons · Quizzes
Salah Tracker	Daily prayer log · Prayer-time reminders · Weekly & monthly reports
Islamic Events	AL QALAM LEARNING MUSABAQA registrations · Ramadan activities · Dhul Hijjah activities · Vacation classes

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Confidential · Al Qalam Learning

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Delivery Plan

We deliver Phase 1 in four working milestones across 16 weeks. Each milestone is a real, usable deliverable — not just an internal checkpoint. You see working features every 4 weeks, and payments are tied to deliverables you can test in your hands.

1

WEEKS 1–4

Design & Foundation

Figma design (wireframes → hi-fi mockups → prototype) · Visual design system · Brand sign-off · Project setup · Auth & role-based access · Student / Teacher / Admin login · Academic calendar · Notification framework

2

WEEKS 5–8

Daily Operations

Attendance · Homework + submissions · Teacher portal core · Push notifications live · Leave application · Workshop registration

3

WEEKS 9–12

Learning & Assessment

Notes & materials · Exam timetable · Marks entry · Report cards · Certificates · Admission form · 1:1 chat

4

WEEKS 13–16

UAT & Go-Live

English + Malayalam polish · Desktop responsive · UAT submission & bug-fix sprint · App Store + Play Store submission · Admin training · Handover & go-live

13

What Happens at Each Milestone

STEP	WHAT AL QALAM DOES	WHAT CODEOX DOES
Milestone demo	Tests the working features on a real device + browser	Deploys a live preview build & walks through every feature
Feedback	Lists any adjustments needed within the locked scope	Incorporates feedback in the next sprint
Sign-off	Confirms the milestone delivery is accepted	Issues invoice for the milestone payment
Payment	Releases milestone payment within 5 working days	Continues to the next milestone

14 Total Investment — Phase 1

ALL-INCLUSIVE · PHASE 1 BUILD

₹ 2,50,000

Two Lakh Fifty Thousand Rupees Only

52+ features · 3 apps · 4 months · 40 / 40 / 20 payment plan · No hidden costs

15 Payment Schedule — 40 / 40 / 20

#	STAGE	WHEN	%	AMOUNT (₹)
1	Advance — kickoff & design sign-off, build begins	On project kickoff	40%	1,00,000
2	UAT Submission — all Phase 1 features built & handed over for client testing	End of Week 12	40%	1,00,000
3	Go-Live — UAT signed off, apps deployed, training complete	End of Week 16	20%	50,000
Total — Phase 1 Build			100%	₹ 2,50,000

16 What's Included & What's Not

✓ Included in ₹ 2,50,000

- ✓ Full Figma design — wireframes, hi-fi mockups, prototype, style guide (mobile + web)
- ✓ Two design revision rounds before development starts
- ✓ Editable Figma file ownership transferred to school
- ✓ iOS app build (React Native)
- ✓ Android app build (React Native)
- ✓ Admin & Teacher Web Portal (Next.js)
- ✓ Student Desktop Web Portal
- ✓ Backend API setup (Node.js + MongoDB on VPS)
- ✓ All 52+ Phase 1 features
- ✓ Bilingual EN + ML
- ✓ Push notifications setup
- ✓ Bug fixes during 4 months of build
- ✓ App Store + Play Store submission
- ✓ Admin & teacher training (one session)
- ✓ Handover documentation
- ✓ 30 days post-launch warranty support

✗ Not Included (Quoted Separately)

- ✗ Phase 2 features (live class integration, online exams, payment gateway, Islamic modules, parent portal)
- ✗ Apple Developer account (USD 99/year — paid by school)
- ✗ Google Play Developer account (USD 25 one-time — paid by school)
- ✗ Cloud VPS rental after launch (~₹800–1,500/month depending on plan & provider — paid by school)
- ✗ S3-compatible object storage for files (typically <₹500/month for one school)
- ✗ Domain name & SSL certificate (Let's Encrypt — free; domain ~₹800/year)
- ✗ Content data entry (notes, materials, student lists) — school's responsibility
- ✗ Custom logo / branding design (if not provided)
- ✗ Annual maintenance contract (optional, after 30-day warranty)
- ✗ Features outside the locked Phase 1 list (handled per the scope-flexibility clause)

Scope Flexibility Clause

Small additions during the project are welcome. **Anything under 1 day of work is absorbed at no extra cost** and batched into the next milestone. Larger changes are quickly estimated — either fit into the current milestone without slipping the deadline, or queued into Phase 2. This keeps the 4-month launch promise intact.

Thank You

جزاكم الله خيراً

For considering CodeOx as your technology partner for Al Qalam Learning. We pray this platform becomes a means of barakah for the students, ustadhs, and families it serves — and a sadaqah jariyah for the founders and supporters of this noble mission.

REACH OUT ANYTIME

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Awaiting your reply — bi-idhnillah, we will begin the moment you give the word.